

Figure 1: Minimal stochastic rupture model and causal synchronization

Code description for GitHub repository (Mevan Rajakaruna)

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Overview

This Python script generates Figure 1 of the manuscript entitled “Feedback polarity controls causal clustering and cascade propensity in stochastic rupture”. The figure comprises four panels (a–d) that illustrate the minimal three-particle, two-bond serial system; tension evolution under displacement control (positive feedback); tension evolution under force control with transient unloading (negative feedback); and the causal synchronization condition based on stress-wave travel time. No data fitting or stochastic simulation is performed; this is a purely deterministic schematic.

1 Software and libraries

The script requires Python version 3.9 or higher. Numerical array operations and sine wave generation for spring visualisation are handled by NumPy (version 1.21 or later). All figure construction, including patches, annotations, and file saving, is performed using Matplotlib (version 3.4 or later).

2 Plotting style

Publication-quality visual styling is enforced through customisation of `matplotlib.rcParams`. The font family is set to sans-serif (Arial or Helvetica), with a base font size of 9 points. Axes labels are 10 points, titles 10 points, tick labels 8 points, and legend text 8 points. The figure is rendered at 300 DPI for screen display and saved at 600 DPI. Line widths are configured as follows: axes lines 0.8 points, tick marks 0.8 points, and main curves 1.5 points. The bounding box is tight with a padding of 0.05 inches.

3 Figure layout

A 4×2 grid is created using `matplotlib.gridspec`. Row 0 occupies the full width and contains panel a. Row 1 is split into two columns: panel b on the left and panel c on the right. Row 2 occupies the full width and contains panel d. Row 3 serves as an empty spacer to preserve clearance below panel d. The height ratios are $[1.00, 1.00, 1.20, 0.25]$, with horizontal spacing of 0.45 and vertical spacing of 0.30.

4 Panel a: Three-particle system (protocol-neutral)

Panel a presents the reference configuration shared by both loading protocols. Three identical particles are positioned at $x = 1, 3, 5$ with $y = 1$ and a radius of 0.4. Two elastic bonds connect particle 1 to particle 2 and particle 2 to particle 3. Each bond is drawn as a straight line with a superimposed sine wave comprising six full periods, mimicking a spring. Particles are lightblue circles with black edges of linewidth 2 and are numbered 1, 2, and 3 inside. Each bond is labeled with the stiffness k and slack length ℓ_0 . The distance between the two rightmost particles is indicated by a double-headed arrow and labeled L . The axis limits are $x \in [-1, 8]$ and $y \in [-1, 3]$, with equal aspect ratio and the axis turned off.

5 Panel b: Displacement control (positive feedback)

Panel b illustrates positive mechanical feedback under displacement control. Pre-rupture tension for each bond is modelled as a linear ramp $T_{\text{pre}}(t) = t$ in arbitrary units. The first rupture time is fixed at $t_1 = 0.8$ for visual demonstration. After the first rupture, the remaining bond experiences a doubled loading rate, described by $T_{\text{post}}(t) = 2(t - t_1) + t_1$ for $t > t_1$. The pre-rupture tension is shown as a blue curve, while the post-rupture tension appears as a red curve. A vertical dashed gray line at t_1 is labeled “First rupture at t_1 ”. A red arrow annotation indicates “Loading rate doubles”. The axis limits are $x \in [0, 2]$ and $y \in [0, 2.5]$, with axes labeled “Time (arb. units)” and “Tension T ”.

6 Panel c: Force control with transient unloading (negative feedback)

Panel c demonstrates negative feedback under force control with transient unloading. The constant target force is set to $F = 1.0$ in arbitrary units. The first rupture occurs at $t_1 = 1.0$. The unloading fraction is $\beta = 0.7$ and the recovery time constant is $\tau_{\text{rec}} = 1.5$. Post-rupture tension follows $T_{\text{post}}(t) = F(1 - \beta e^{-(t-t_1)/\tau_{\text{rec}}})$ for $t > t_1$. Pre-rupture tension is constant at F and shown as a horizontal blue line. The post-rupture tension is plotted in red. A vertical dashed gray line marks t_1 , and a horizontal dotted red line indicates the immediate post-rupture level $F(1 - \beta)$. A red arrow annotation displays “Unloading $\beta = 0.7$ ”. Axis limits are $x \in [0, 4]$ and $y \in [0, 1.2]$.

7 Panel d: Causal synchronization schematic

Panel d provides an operational definition of near-simultaneous rupture based on causality. Two circular failure sites are drawn at coordinates $(2, 2)$ and $(6, 2)$, each with radius 0.5. The first failure site is colored lightcoral with a red edge and annotated t_1 . The second site is lightblue with a blue edge and annotated $t_2 = t_1 + \tau$, where τ is the delay between failures. The separation between sites is $L = 4$ units. Stress waves propagate at speed c_s , represented by concentric dashed orange circles expanding from the first site with radii $r = 1.0, 2.2, 3.4$. An orange arrow labeled “Stress wave speed c_s ” points from the first site toward the second. A double-headed black arrow between the sites is labeled L , and the material causal time $t_{\text{mat}} = L/c_s$ is displayed beneath it. A light yellow

box summarises the synchronization condition: if $\tau < t_{\text{mat}}$ the failures are considered synchronized; if $\tau > t_{\text{mat}}$ they are independent.

8 Output files

The script saves the figure in three formats. An Encapsulated PostScript file is written as `fig1_schematic.eps`. A high-resolution PNG file at 600 DPI is written as `fig1_schematic.png`. A Portable Document Format file is written as `fig1_schematic.pdf`. All files are saved in the current working directory. The figure is also displayed interactively via `plt.show()`, and the script prints a confirmation message upon completion.

9 Reproducibility

No random numbers are used; the figure is entirely deterministic. No external data files are required. To regenerate the figure exactly, the script must be executed in a Python environment with the specified library versions. The script prints “Figure 1 saved as `fig1_schematic.eps`, `.png`, `.pdf`” upon successful completion.

10 Relation to the manuscript

Panel a corresponds to Figure 1a in the paper, showing the three-particle series. Panel b reproduces the displacement-control tension evolution from Equation (5) and Figure 1b. Panel c reproduces the force-control transient unloading from Equation (6) and Figure 1c. Panel d reproduces the causal synchronization definition from Equations (1)–(2) and Figure 1d.

License and citation

This code is part of the research by M. Rajakaruna and H. Rajakaruna. If you use this code or the generated figure, please cite the associated manuscript as described in the `README.md` file in the repository.